

MASTER'S THESIS DEFENCE

NupM1-mediated regulation of transporter systems and drug response in *Leishmania donovani*

Presented by Harman

MS in Pharmaceutical Sciences

Thesis mentor: Dr. Nicola Carter



Family is everything

About Me

Harman Kaur PharmD



Class of 2025

Started Pharmacy school the same year my daughter started her Kindergarten



Bowling Fan

Love hitting the lanes with family and friends.



Binge Watcher

K-dramas, sci-fi, Supernatural & horror shows



Big Dreamer

Do something meaningful to help humanity

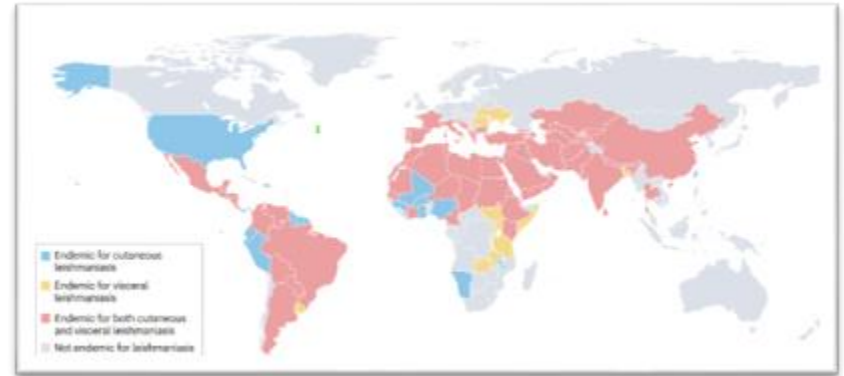


My Motto

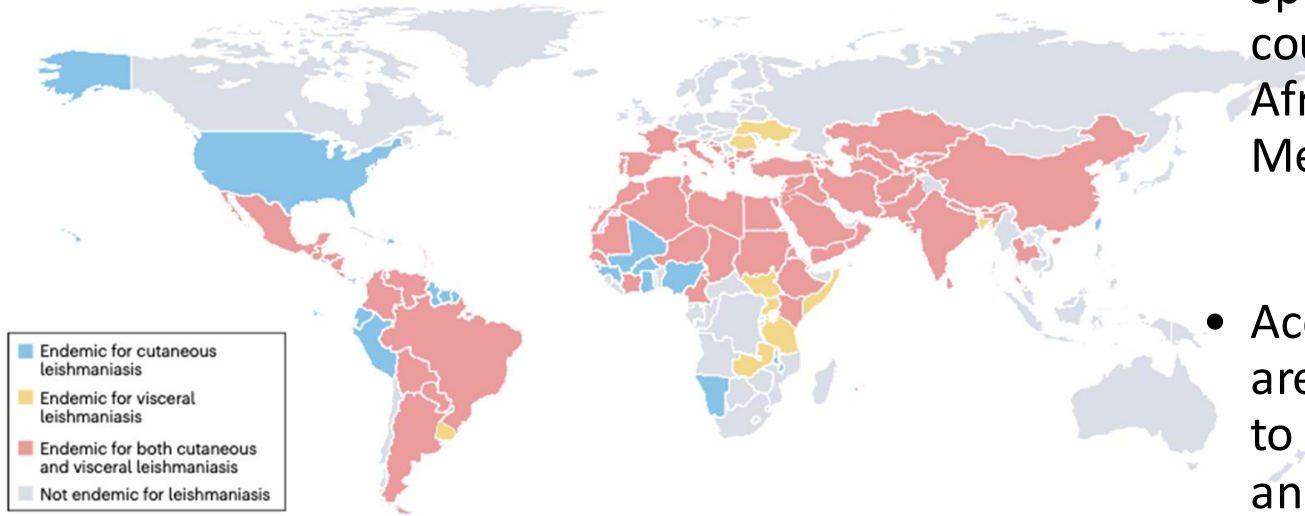
Dream big, keep going, and never give up!

Background: Leishmaniasis and Treatment Challenges

- Leishmaniasis, caused by *Leishmania* spp., is a significant global health concern and remains a major neglected tropical disease.
- Therapeutic options are limited and impeded by toxicity, cost and emerging drug resistance.
- Better understanding of parasite biology can inform new therapeutic approaches
- Parasite transporter systems are attractive targets because govern drug uptake and efflux and several implicated in drug resistance mechanisms.

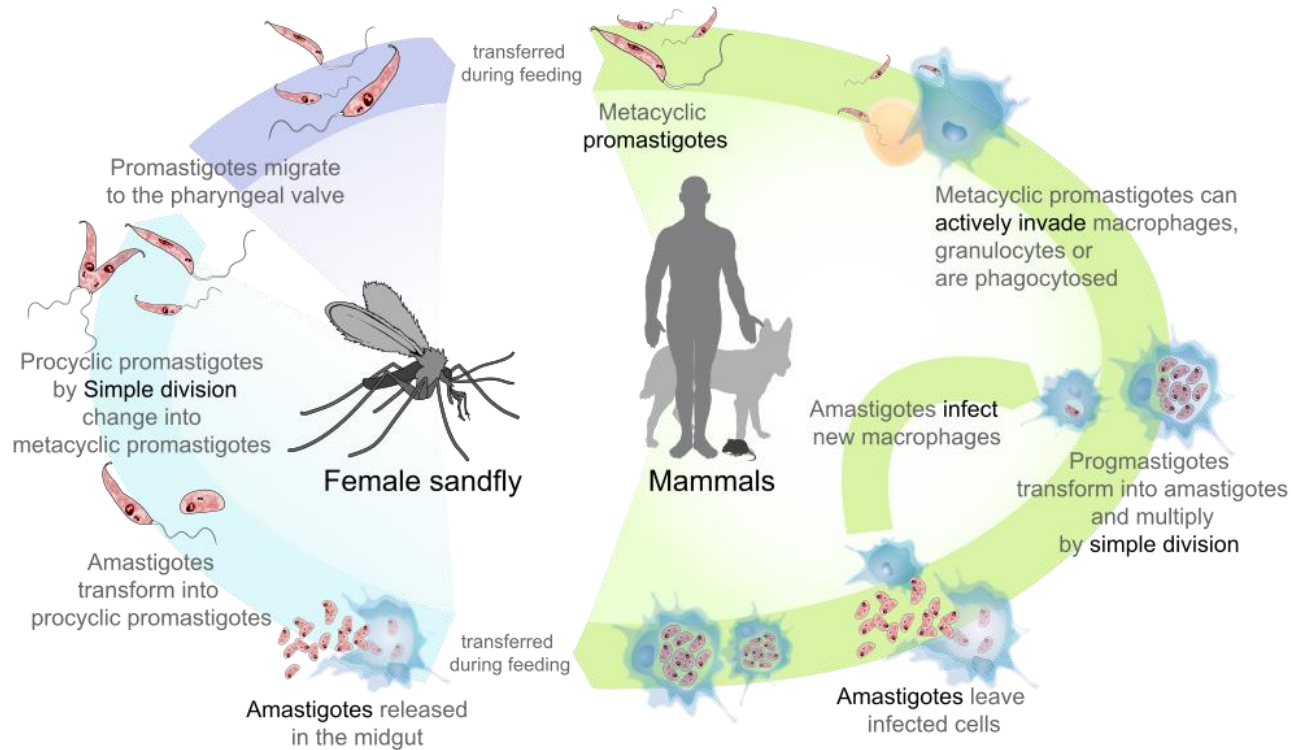


Prevalence of Leishmaniasis



- Spreading across nearly 90 countries including Asia, Africa, and the Mediterranean
- According to WHO, there are an estimated 700,000 to 1 million new cases annually

Digenetic Life Cycle of *Leishmania*



Cutaneous leishmaniasis

Most common form of the disease.

Species:

L. tropica, *L. major*, *L. aethiopica*, *L. infantum*, *L. donovani*, *L. mexicana*, *L. amazonensis*, *L. venezuelensis*

Incubation period:

2 weeks to several months (rarely up to 3 years)

Symptoms:

Papule at the site of Sandfly bite and increase in size

Crusts

Ulcerates



<https://www.cmaj.ca/content/188/14/1034>

Resource:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3202701/>

Visceral leishmaniasis

- Can lead to death if left untreated

Species:

L donovani, *L infantum*, *L chagasi*,

Incubation period:

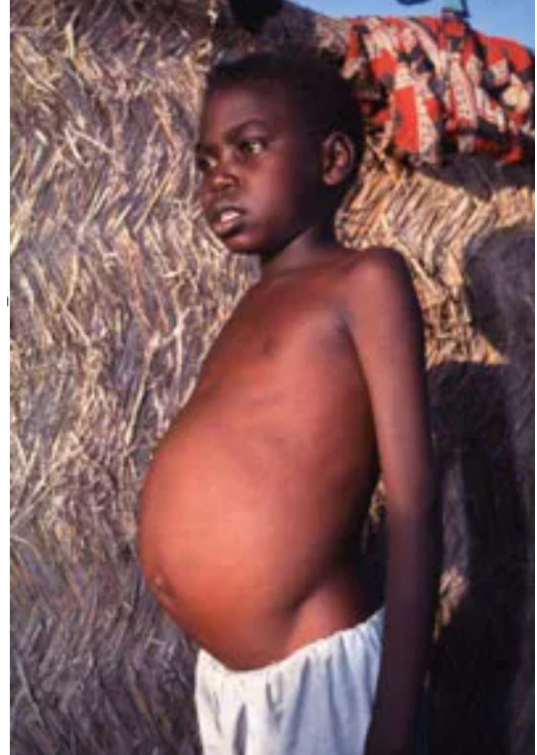
10 days to 34 months

Symptoms:

Fever, Weight loss, Enlargement of Spleen than liver,
Lymphadenopathy, Pancytopenia,
Hypergammaglobulinemia skin pigmentation (PKDL)

Resources:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3202701/>



https://warwick.ac.uk/fac/cross_fac/zeeman_institute/new_research/combating_disease/visceralleishmaniasis/

Comparison of Treatments for Leishmaniasis

Drug	Class	MOA	Route	Indications	Side Effects
Amphotericin B Liposomal	Antifungal	Binds to Parasite Ergosterol and Pore formation	Parenteral	Second Line or refractory cases	Nephrotoxicity Infusion reaction
Miltefosine	Alkyl-phosphocholine	Disrupt lipid metabolism and Mitochondrial function	Oral	Various Species	Gastrointestinal upset Teratogenicity
Pentamidine	Antiprotozoal	Interference of DNA synthesis and Mitochondrial topoisomerase II	Parenteral	<i>L. Guyanesis</i> infection	Nephrotoxicity Hypotension Hyperglycemia
Antimony (Antimoniate Meglumine) & (Sodium Stibogluconate)	Pentavalent Antimony	Prodrug Interfere DNA repair and disrupt cell redox balance, Inhibit Trypanothione reductase and DNA Topoisomerase I	Parenteral	First Line for most species	Cardiotoxicity Hepatotoxicity Pancreatitis
Paromomycin	Aminoglycoside ABO	Inhibit Protein synthesis	Oral, Topical	First Line for Cutaneous Leishmaniasis	GI upset, Nausea, Hepatotoxicity, Skin sensitivity or rash

Table: Drugs use in treatment of Leishmaniasis and their MOA

Membrane protein trafficking pathway in *Leishmania* parasite

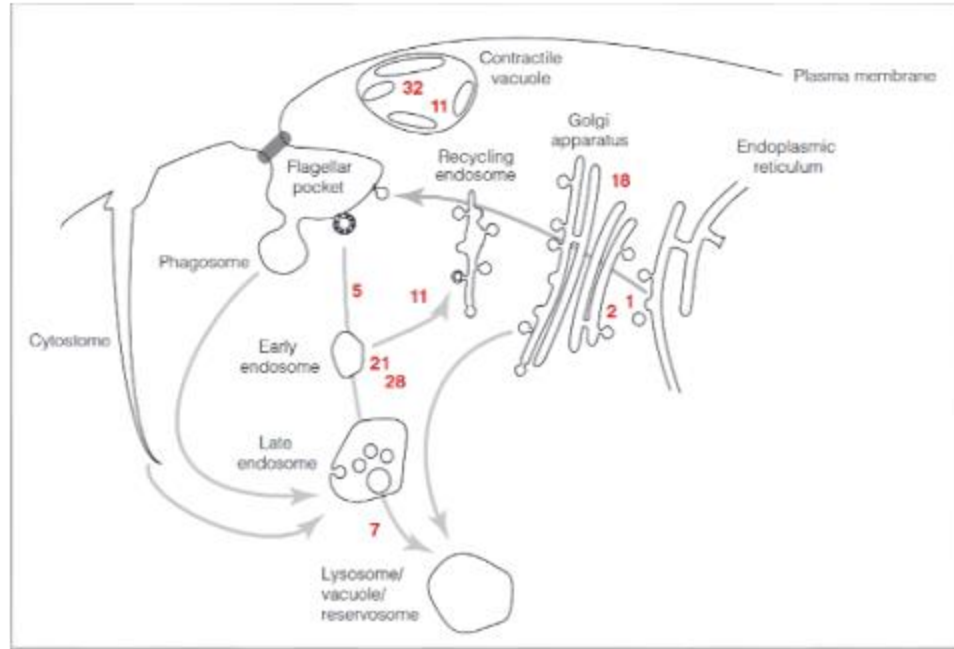
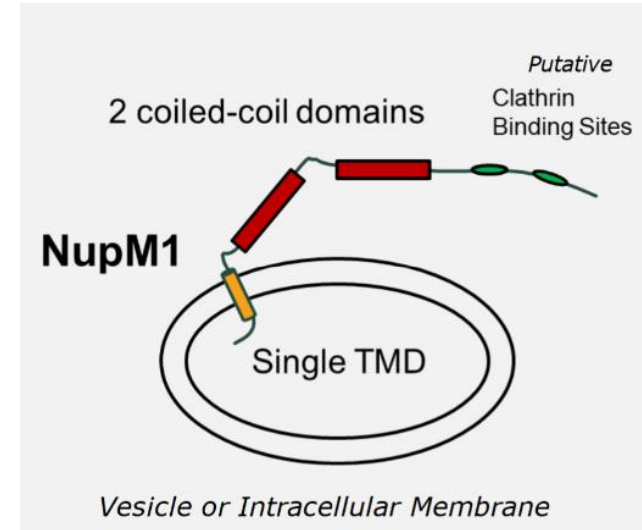


Figure 9: Trafficking pathways of kinetoplasts and several Rab GTPase locations are shown in red as an example (Venkatesh et al., 2018)

NupM1: Discovery and Protein Architecture

- Identified within the Detke lab as a toxic nucleoside resistance determinant or TOR (*name subsequently changed to NupM1*).
- 53.1 kDa, 478 amino acid residues; no known orthologs outside *Leishmania* and related species.
- Bioinformatic analyses: Single N-terminal transmembrane domain and two predicted coiled-coil domains Localizes to Golgi, plasma membrane and multivesicular tubule/lysosome
- Overexpression NupM1 lowers level of surface LdNT1 protein, redirecting it to multivesicular tubule = parasite lysosome.



Research Problem and Knowledge Gap

KNOWN

NupM1 overexpression alters drug sensitivity and resistance phenotypes to toxic purine drugs by altering the abundance of purine transporters at the parasite surface.

UNKNOWN

1. Does NupM1 alter the abundance and function of other transporters within *Leishmania*?
2. How are the interactions of NupM1 with transport proteins (or accessory trafficking proteins) are governed?

Table 8: Purine transporters and cell surface activities are upregulated by NT1-3 at protein level and were verified at 96 hour starved parasites. (data from Carter lab)

Main Phosphosites Identified in NupM1	Average fold increase of phosphorylation in purine-starved cells at 96 h
eGAVsEDEFAGDsSDTSR*	4.30
sAsVAsSAPRPLEDVLGTDcESYPFLmNVER	0.22
dVEHNVALSGFFNtPTNNNMASVDTAsR	-2.06
vDTQRPSPHPTALSANTSR	1.05
eRNsPEQLGSSSSGEAAVER	1.45
ttSTVAYGR	1.90

The effect of purine starvation on the phosphorylation of the NupM1 protein experiment

Phosphosites Identified in NupM1	Average fold increase of phosphorylation in purine-starved cells at 96 h
eGAVsEDEFAGDsSDTSR*	4.30
sAsVAsSAPRPLEDVLGTDc ESYPFLmNVER	0.22
dVEHNVALSGFFNtPTNNN MASVDTAsR	-2.06
vDTQRPsPHPTALSANTSR	1.05
eRNsPEQLGSSSSGEAAV ER	1.45
ttSTVAYGR	1.90

Does phosphorylation of NupM1 at serine 136 and 144 alters its conformation and regulatory function on purine transporter abundance?

Specific Aims

AIM 1

Is NupM1 a global regulator of transporter abundance?

Determine whether NupM1 overexpression alters sensitivity to antileishmanial drugs and cytotoxins by modulating their uptake into the parasite.

AIM 2

What impact on NupM1 function does phosphorylation confer?

Test whether phosphorylation at serine 136 and serine 144 modulates NupM1 function and consequently LdNT2 activity at the parasite surface.

SA1 Experimental System and Cell Lines

WT

Wild-type *L. donovani*

Baseline drug-response reference

A2

NupM1-overexpressing line

Tests the effect of elevated NupM1 on drug response

Positive Control

Formycin B: overexpression of NupM1 should confer resistance to this drug by decreasing LdNT2 surface expression/activity.

Experimental Conditions

Test for resistance to various drugs and cytotoxins that are known to use specific transporter proteins in *Leishmania* in the A2 NupM1 overexpression line compared to WT.

Aim 1 Methods: Drug Sensitivity Assays



- WT and A2 cells were treated with Formycin B, Pentamidine, G418, Miltefosine, antimony, puromycin, DFMO and methotrexate.
- EC50 values were derived from dose-response curves for each line.
- Differences were assessed statistically.

Aim 1 Key Results: Altered Drug Sensitivity

- A2 cells were more sensitive than WT to Pentamidine and G418.
- Consistent with reduced activity or abundance of efflux-associated transporters.
- Formycin B resistance indicates altered nucleoside transporter-mediated uptake.

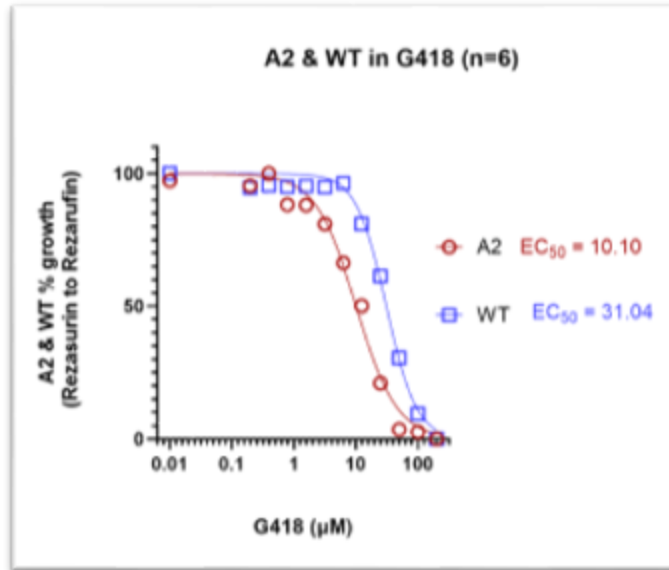
Table: Data table for EC₅₀ values_analysis

Drug	EC ₅₀ (μM) ± SD of A2	EC ₅₀ (μM) ± SD of WT	Two tail T-Test
G418 (n=6)	13.25 ± 4.54	30.25 ± 11.87	0.02
Pentamidine (n=6)	1.52 ± 0.86	3.33 ± 0.93	0.00
Formycin B (n=6)	0.21 ± 0.11	0.02 ± 0.00	0.01
Miltefosine (n=3)	0.63 ± 0.23	0.63 ± 0.18	NS
Antimony (n=3)	44.83 ± 2.91	35.69 ± 9.44	NS
Puromycin (n=6)	10.85 ± 2.30	10.69 ± 2.07	NS
DFMO (n=5)	41.65 ± 48.09	39.29 ± 24.54	NS
Methotrexate (n=2)	15.58 ± 15.05	0.83 ± 0.61	NS

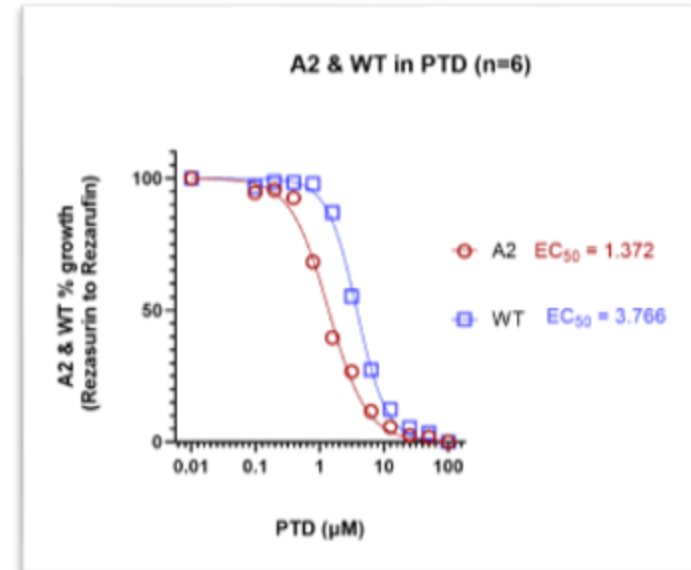
Note: NS - Not Significant; EC₅₀ - Median Effective Concentration, SD – Standard Deviation

G418 and Pentamidine Curves

G418 EC₅₀ Values



Pentamidine EC₅₀ Values



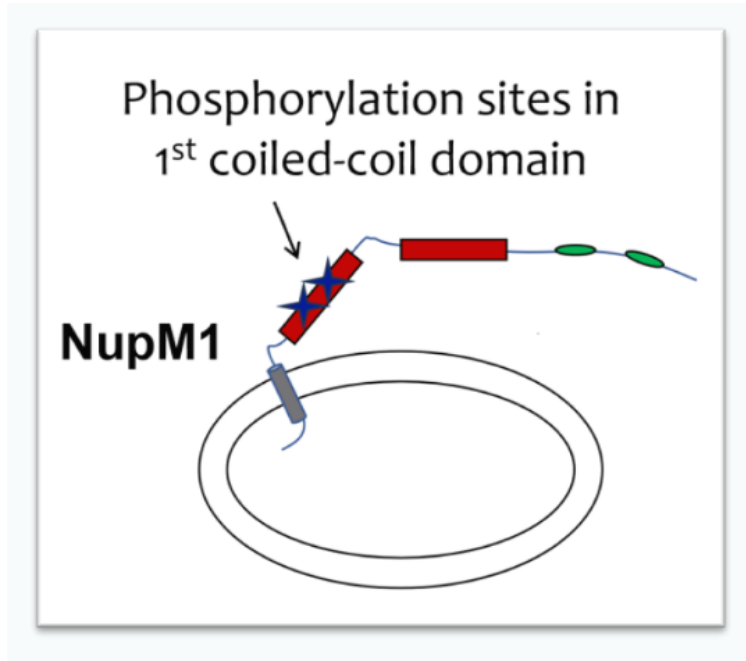
Aim 1 Interpretation: Selective Transporter Effects

Table: Known transporters of Antileishmanial drugs and other toxic analogs.

Drug Name	Known Transporter protein
Miltefosine	LdMT (Garcia-Salcedo et al., 2016)
Allopurinol	LdNT3 & LdNT4 (Ortiz et al., 2007)
Formycin B	LdNT2 (Carter et al., 2000; Iovannisci et al., 1984)
Tubercidin	LdNT1 (Iovannisci et al., 1984; Vasudevan et al., 1998)
Antimony	AQP1 (ABCC3 and LABC2) (Frézard et al., 2014)
Puromycin	PGP (Ldmdr1) (Theile et al., 2010)
Methotrexate	Folate Transporter (Dean et al., 2014)
DFMO	LdAAP24? (Carter et al., 2022)
G418 (Geneticin Sulphate)	Phosphorylation gene (Neo [®]) (Theile et al., 2010)
Pentamidine	ABCC7/PRP1 (Garcia-Salcedo et al., 2016)

Increased sensitivity	Resistance / control	No clear effect
Pentamidine, G418	Formycin B	Miltefosine, Antimony, Puromycin
Efflux-associated routes appear affected	Nucleoside transporter uptake altered	DFMO and Methotrexate gave no clear dose-response

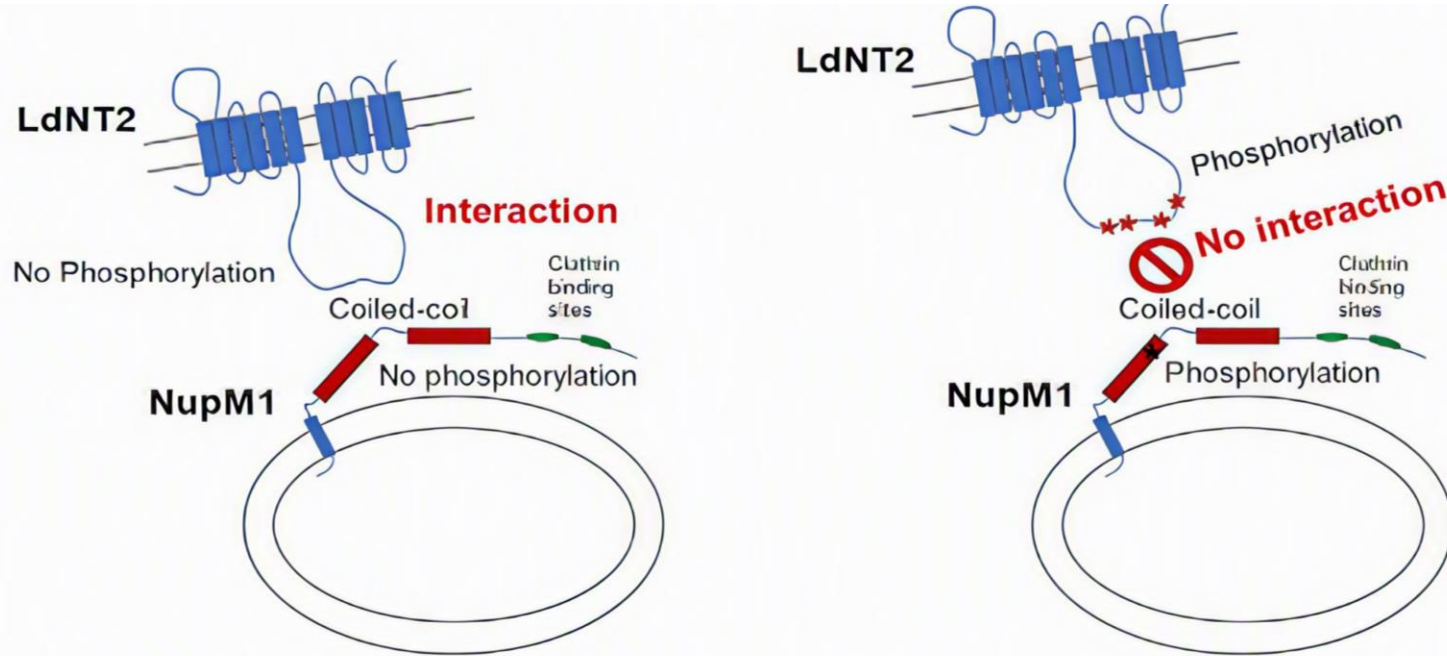
Aim 2 Rationale: The Role of Phosphorylation on NupM1 Regulatory Function



*Proposed membrane-associated domain architecture of NupM1.
Created by Carter, Nicola.*

- NupM1 contains two predicted coiled-coil regions and the phosphosites at Ser136 and Ser144 are predicted to localize to the first coiled-coil domain.
- Purine deprivation increases the cell surface abundance of LdNT1-3 and increases the phosphorylation of NupM1 with no impact on its protein abundance.
- Hypothesis to be tested: phosphorylation of NupM1 at serine 136 and 144 reduces its ability to interact with the purine transporters and send them to the parasite lysosome.

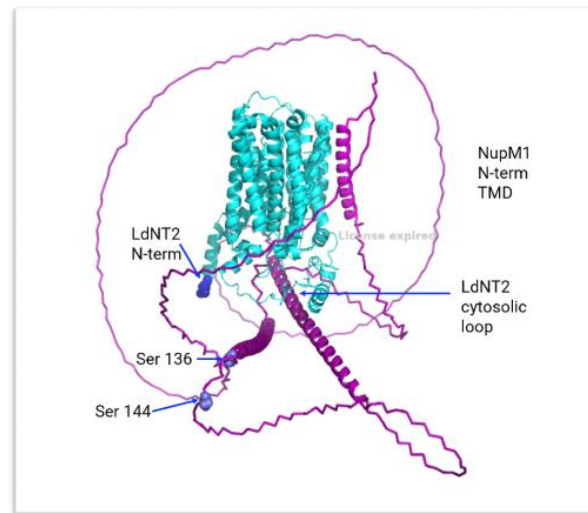
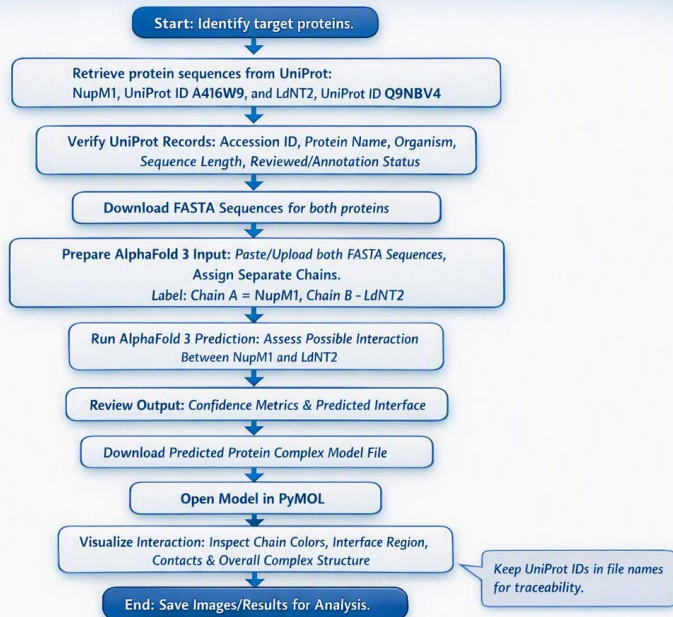
Hypothesis: Phosphorylation Impedes NupM1 Binding



In Silico Analysis to look at the effect of phosphorylation on NupM1 protein and its interaction with LdNT2



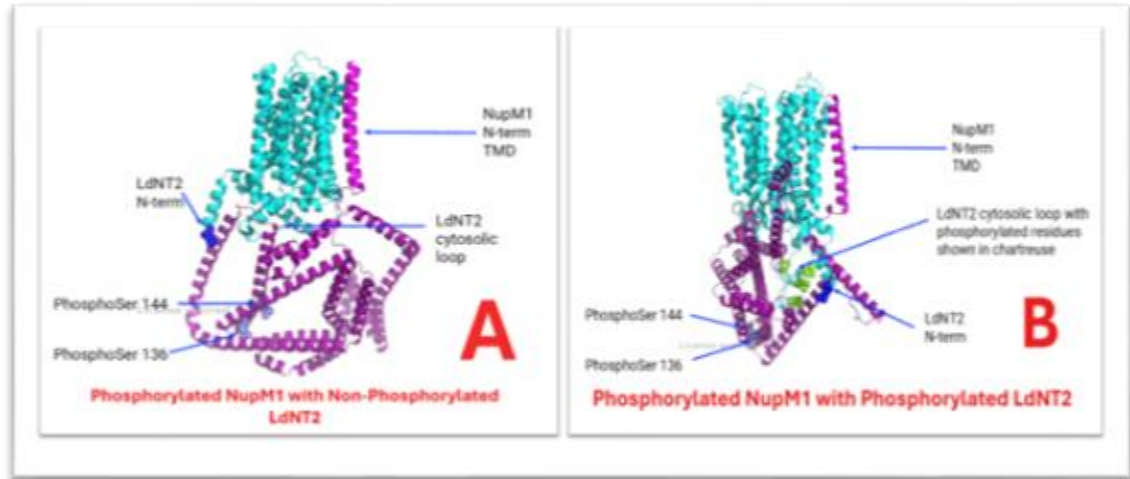
Detailed UniProt—AlphaFold 3—PyMOL Workflow



This model predicts minimal interaction between NupM1 and LdNT2 when they are not phosphorylated

In Silico Analysis to look at the effect of phosphorylation on NupM1 protein and its interaction with LdNT2

- When the phosphoserine at positions 136 and 144 are included in the model the conformation of NupM1 changes dramatically (A)
- Phosphorylated NupM1 adopted a more ordered conformation and shifted relative to LdNT2.
- When phospho-serine at 260 and 263, and phospho-threonine at 268 and 283 within the LdNT2 sequence were modeled, the large cytosolic loop (in green) of LdNT2 was surrounded by phosphorylated NupM1.
- *This model predicts that the interaction between NupM1 and LdNT2 is increased when they are phosphorylated*



A: phosphorylated NupM1 with Non-phosphorylated LdNT2. B: phosphorylated NupM1 with phosphorylated LdNT2. Created via AlphaFold 3.

IMPORTANT: Computational predictions generating testable hypotheses, not experimental structures.

Aim 2 Method for Generating NupM1 Phosphosite Mutants

Primer design

Targeting Ser136 and Ser144

QuikChange mutagenesis

Site-directed substitution

Clone screening

PCR, MAMA-PCR, sequencing

Transfection

Into *Leishmania* for analysis

Phosphomimetic: S136D, S144D

Aspartate mimics the phosphate charge, simulating constitutive phosphorylation.

Phospho-null: S136A, S144A

Alanine cannot be phosphorylated, blocking regulation at that residue.

Aim 2 Workflow: Mutagenesis of NupM1-GFP

Design primers

Mutagenic primers for Ser136 and Ser144

Mutagenize

QuikChange on the plasmid pRP-H-SAT⁺-DiCre-NupM1-GFP

Propagate and select

Transform competent *E. coli*; clones grown overnight with Carbenicillin

Screen and confirm

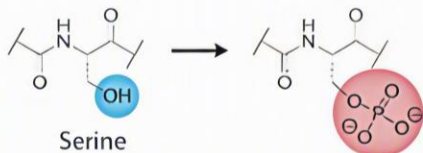
PCR, MAMA-PCR, sequencing

Transfect cells

Into *L. donovani*; select clonal lines with NTC

Phosphomimetic: S136D, S144D

Constitutively phosphorylated NupM1-GFP.



Phospho-null: S136A, S144A

NupM1-GFP locked in the unphosphorylated state.



Lanes, left to right

- 1 — DNA ladder
- 2 — S136A 3 — S136D
- 4 — S144A 5 — S144D
- All four mutagenesis reactions yielded a product that was apparent after the Dpn1 digest.
- 2 μ L of the PCR was transformed into chemically competent *E. coli* and 10 clones per DNA were selected for analysis by MAMA-PCR screening.

MAMA-PCR: Principle Behind Mutant Screening

Allele-specific primers

Amplify only the mutant sequence

3' mismatch discrimination

Taq cannot extend a mismatched base

T_m-paired primer design

Matched melting temperatures

PCR

At specified annealing temperatures

Why it is used

Mismatch amplification mutation assay: rapid, low-cost mutant screening before sequencing.

Applied here

Applied to Ser136 and Ser144 clones ahead of confirmation.

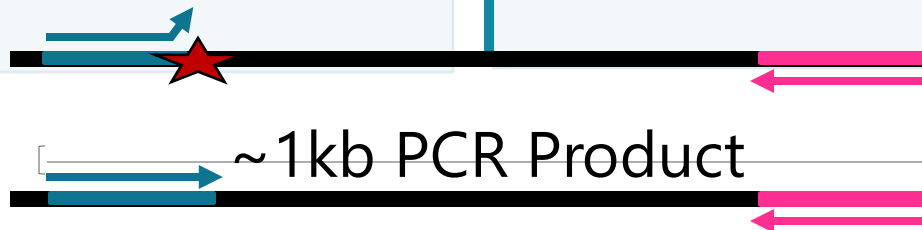
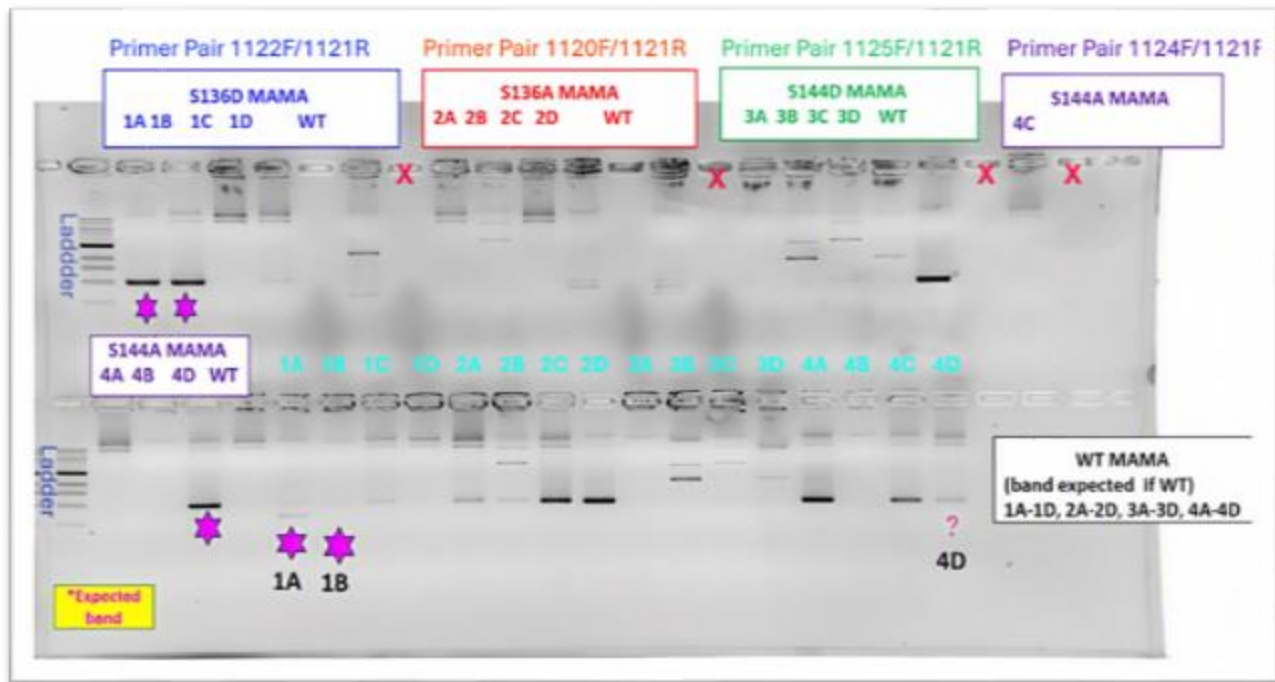


Figure illustrating the principle of Mismatch Amplification Mutant Analysis PCR.

Aim 2 Results: MAMA-PCR Screening Gel



MAMA-PCR panels for S136D, S136A, S144D and S144A; positive clones 1A, 1B (Ser136D) and 4D (Ser144A) highlighted. These candidate positive clones were identified as carrying the correct mutation and were submitted to Plasmidosaurus for DNA sequencing which confirmed the intended mutation and no additional secondary mutations.

Sequence Verification and Plasmid Linearization

Whole-plasmid sequencing

Plasmid DNA verified by Plasmidsaurus before nucleofection

Oxford Nanopore long-read

Enables full-length plasmid sequence verification

AvrII restriction digest

Confirmed plasmid DNA linearized before transfection

Manufacturer conditions

AvrII-compatible buffer and recommended incubation

Why it matters

Full-length reads confirm the intended mutation and rule out unintended changes elsewhere in the construct.

Endpoint

Digestion run to completion, giving fully linearized DNA ready for nucleofection.

Aim 2 Approach: Expression and EC50 Measurement

1. Nucleofection

Sequence-confirmed,
AvrII-linearized mutant
and WT NupM1-GFP DNA
into WT cells

2. Select clones

Nourseothricin selection,
then limiting dilution

3. Verify expression

Quantitative Western:
anti-GFP, tubulin loading
control

[How to Achieve Quantitative
Western Blot Data: a Step-by-Step
Guide](#)

4. Determine EC50

Dose-response curves
and EC50 for every line

Question

Do S→D or S→A mutants shift EC50 for Formycin B,
G418 or Pentamidine?

Control

Miltefosine serves as the control drug.

Overall Conclusions

01

NupM1 overexpression produces selective, not global, changes in drug sensitivity.

02

NupM1 overexpression cells showed statistically significant increased sensitivity to Pentamidine and G418 and resistance to Formycin B.

03

In silico modelling supports phosphorylation as a regulator of NupM1 conformation and likely its functional interactions with LdNT2.

04

Though we do not yet know how phosphorylation impacts NupM1 function, we propose that it is a phosphorylation-regulated trafficking factor controlling surface transporter abundance.

Limitations in SA2 Study

- AlphaFold models while intriguing are predictive and require further experimental validation.
- Only two of the four planned phosphosite constructs could be recovered for preliminary testing.
- Further analyses of the effect of these mutations will rely on Formycin B sensitivity testing which is an indirect readout of LdNT2 localization.

Future Directions

1

Complete the mutant panel

Generate and sequence-confirm all four mutants: S136A, S136D, S144A, S144D.

2

Strengthen the phenotype

Conduct Formycin B EC50 assays across biological replicates.

3

Localize the transporter

Assess LdNT2 localization by fluorescent epitope tagging, fluorescence microscopy to visualize, or surface labelling to quantify.

4

Prove the interaction

Test NupM1-LdNT2 binding by co-immunoprecipitation or proximity labelling.

Key References

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Acknowledgements

SUPERVISION

Mentor: Dr. Carter

Thesis committee: Dr. Stamper, and Dr. Roberts

LAB

Jennifer Chu

Heena Patel

Jon Taylor

Elham Yaaghubi

INSTITUTION

Pacific University Pharmacy Department, funding sources, and technical support

PERSONAL

Family

Thank you.

Questions
